

Competitor Price Intelligence Brief

Task T44 | AI Text Analytics | 30 Marks

Objective: analyse price gaps against competitors by category and platform, create a verified weekly management brief, identify overpriced and underpriced products, and recommend evidence-based pricing actions.

KPI	Result
Source rows	400
Comparable observations	100
Overpriced	49
Underpriced	51
Average gap	-0.7%
Same-platform coverage	58%
Beyond ±5% review threshold	65

Executive Summary

Across 100 comparable observations, Our Brand is on average 0.7% below the three-competitor benchmark, so the portfolio is broadly near market overall. However, the average absolute gap is 7.8%, showing that individual products can still diverge meaningfully even when the total average is small.

Fashion has the highest category-level average gap at +2.3%, while Grocery is lowest at -2.3%. In the latest week (16 Feb 2026), 3 observations are overpriced and 7 are underpriced, with an average gap of -5.4%.

The pricing recommendation is not to automate price changes. A ±5% gap is used only as a human-review trigger. Platform comparability, stock status, current promotions, product positioning and margin must be checked before action.

1. Methodology and Working Build

The provided Price Tracker contains 400 synthetic observations. The data follows a repeating seller structure: Our Brand, Competitor A, Competitor B and Competitor C. The analysis converts these 400 rows into 100 comparable four-seller observations without collapsing repeated product/week combinations.

Price-gap calculation

- Competitor benchmark = arithmetic mean of Competitor A, Competitor B and Competitor C for the same observation.
- Price gap (INR) = Our Brand selling price - competitor benchmark.
- Price gap % = price gap / competitor benchmark $\times$ 100.
- Positive gap = Our Brand is more expensive; negative gap = Our Brand is cheaper.
- Same-platform competitor coverage is calculated separately because cross-platform comparisons are less directly comparable.

Working build

A Streamlit application was developed in Python. The app loads the bundled T44 workbook by default, while an upload control allows another workbook with the same schema to be analysed. Deterministic calculations are performed in Python rather than delegated to an LLM.

- Executive dashboard with KPI cards and category signals.
- Threshold slider and category/platform filters.
- Action Centre for the largest review signals.
- Product, category, platform and weekly analysis.
- Verified weekly brief and LLM prompt workflow.
- Verification checks and downloadable multi-sheet Excel output.

Why AI is used this way

The LLM is used for interpretation, drafting and critique. It is not the source of benchmark arithmetic. This separation reduces hallucination risk: numbers are computed first, a management brief is then drafted from verified metrics, and numeric claims are checked again before acceptance.

2. Findings

Category-level position

Category	Obs.	Avg gap	Same-platform
Fashion	15	+2.3%	67%
Electronics	15	-0.4%	40%
Beauty	21	-0.5%	67%
Home & Kitchen	20	-0.9%	55%
Grocery	29	-2.3%	59%

Fashion is the only category with a clearly positive average gap (+2.3%), meaning it is more expensive than the three-competitor benchmark on average. Grocery is lowest at -2.3%. Electronics, Beauty and Home & Kitchen are close to the benchmark in aggregate.

Our Brand platform position

Our Brand platform	Obs.	Avg gap	Same-platform
Quick-commerce app	32	+0.9%	50%
Own website	31	-1.0%	65%
Marketplace	37	-1.7%	59%

Quick-commerce observations average +0.9%, Own website -1.0% and Marketplace -1.7%. These are descriptive signals rather than causal platform effects because competitor platform, stock and promotion status can differ.

Persistent product signals

Product	Obs.	Avg gap	Signal	Evidence
Running Shoes	2	+12.8%	Persistent premium	Moderate evidence
Denim Jeans	2	-11.2%	Persistent discount	Moderate evidence
Face Serum	3	-10.0%	Persistent discount	Moderate evidence
Mixer Grinder	2	-9.9%	Persistent discount	Moderate evidence
USB-C Charger 65W	4	-7.5%	Persistent discount	Higher evidence
Formal Shirt	3	+5.5%	Persistent premium	Moderate evidence
Basmati Rice 5kg	2	+5.5%	Persistent premium	Moderate evidence

Running Shoes is the strongest persistent premium signal at +12.8% across two observations. Formal Shirt averages +5.5% across three observations. On the discount side, Face Serum averages -10.0% across three observations and USB-C Charger 65W averages -7.5% across four observations. These products deserve review, but small sample sizes and platform differences should be considered.

3. Latest Weekly Brief and Recommendations

Latest week analysed: 16 February 2026. There are 10 comparable observations: 3 overpriced and 7 underpriced. The average gap is -5.4% and same-platform coverage is 70%.

Product	Category	Our price	Benchmark	Gap	Suggested review
Formal Shirt	Fashion	₹1,697	₹1,573	+7.9%	Review premium / promo
Cold-pressed Oil 1L	Grocery	₹405	₹432	-6.3%	Check margin headroom
USB-C Charger 65W	Electronics	₹1,326	₹1,464	-9.4%	Check margin headroom
Cold-pressed Oil 1L	Grocery	₹305	₹350	-12.8%	Check margin headroom
Beard Trimmer	Beauty	₹1,353	₹1,554	-13.0%	Check margin headroom
Pressure Cooker 5L	Home & Kitchen	₹1,315	₹1,567	-16.1%	Check margin headroom

Recommended actions

- Formal Shirt: latest-week gap is +7.9%. Review the price or promotion; retain the premium only if product positioning or margin strategy supports it.
- Pressure Cooker 5L: latest-week gap is -16.1%. Check margin headroom before maintaining such a large discount.
- Beard Trimmer (Marketplace): latest-week gap is -13.0%. Same-platform evidence is available, so this is a stronger candidate for margin review.
- Cold-pressed Oil 1L: one latest-week observation is -12.8%. Validate the repeated product observations separately rather than averaging away variation.
- USB-C Charger 65W: latest-week gap is -9.4% and the product also shows a persistent -7.5% average gap across four observations; this is one of the stronger repeat discount signals.

Management should avoid automatic repricing. A price move should follow a commercial check of stock status, promotions, competitor platform, customer value proposition and margin impact.

4. Verification, Responsible AI, Limitations and Reflection

Verification performed

Check	Status	Result
Source row count	PASS	Found 400 rows; task brief says 400.
Complete observations	PASS	Built 100 complete Our Brand + 3 competitor observations.
Expected seller set	PASS	Sellers: ['Competitor A', 'Competitor B', 'Competitor C', 'Our Brand']
No missing selling prices	PASS	Missing/non-numeric: 0
Three-competitor benchmark	PASS	Benchmark equals mean(Competitor A, B, C) for every observation.
Gap formula	PASS	$\text{price_gap_inr} = \text{our_price_inr} - \text{competitor_avg_price_inr}$ for every observation.
Gap % formula	PASS	$\text{price_gap_pct} = \text{price_gap_inr} / \text{competitor_avg_price_inr} \times 100$.
Three competitors per observation	PASS	Each observation is traceable to exactly three competitor source rows.
Same-platform logic	PASS	Same-platform count is restricted to 0–3 competitors.

The project test script also passed end to end: 400 source rows were converted into 100 observations; the three-competitor benchmark, gap formulas and Excel export were verified.

Responsible AI and limitations

- All supplied data is synthetic and created for teaching; no real personal or confidential business data was sent to a public AI tool.
- Every business number in the final brief comes from deterministic calculations and verification checks, not from an LLM guess.
- Same-platform competitor coverage is 58%, so some comparisons are cross-platform and require caution.
- Availability, promotions, fulfilment, brand positioning and margin are not fully captured by a simple price-gap metric.
- Several products have only one to three observations, so persistence claims are stronger for products with more repeated observations.

Personal reflection

The first approach focused too much on a static workbook. Reviewing the rubric changed the direction: the project was rebuilt as a working Streamlit tool, tested locally, and then improved after reviewing the first app output. The second version made the business decision more visible through executive KPIs, an Action Centre and explicit verification.

The most useful learning was that generative AI should not be treated as the calculator. The stronger workflow is: compute and verify first, ask the LLM to communicate the findings second, then re-check every numeric claim. This makes the AI useful without giving it authority over the underlying business facts.

Final recommendation: use the dashboard as a weekly review tool. Prioritise repeated gaps above $\pm 5\%$, starting with products that also have same-platform competitor evidence, and require a human commercial check before any pricing decision.